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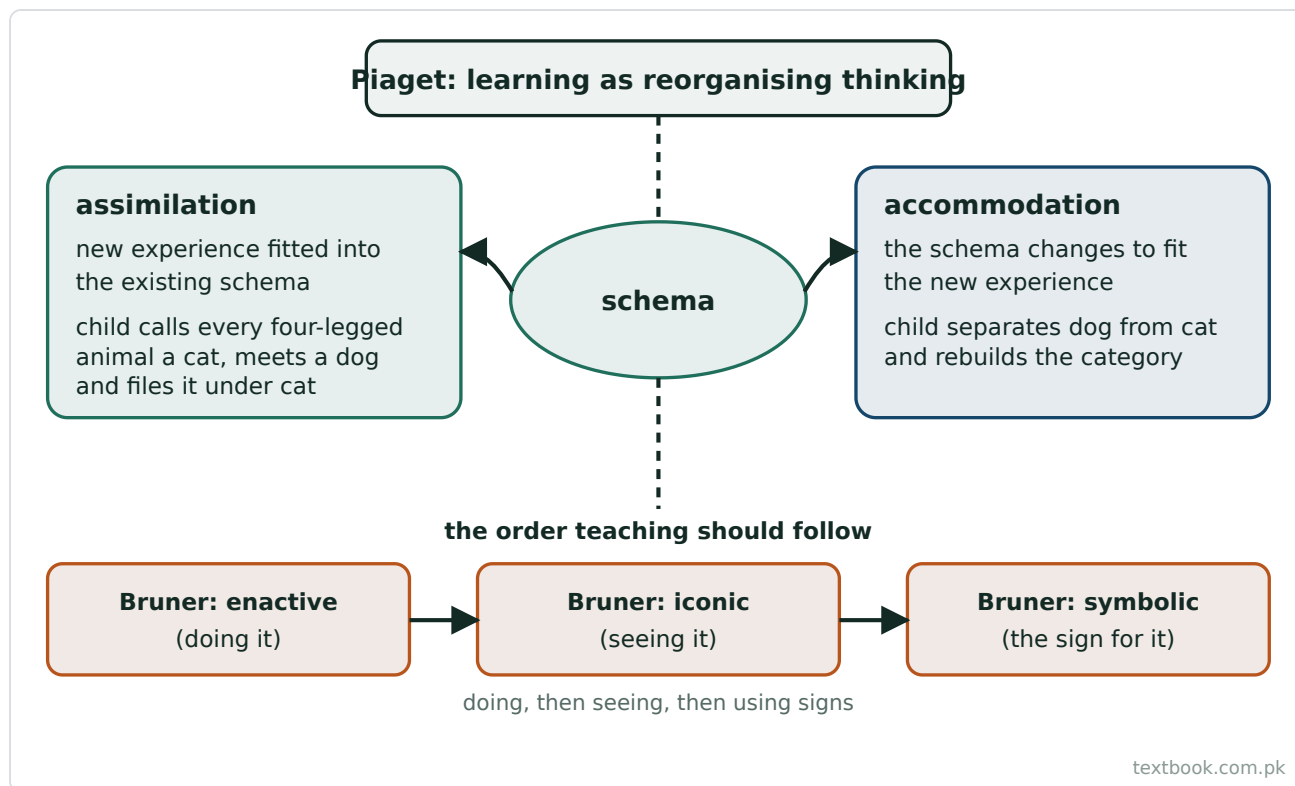
Topic 3.2 - The cognitive turn: Piaget and Bruner

A real classroom situation

Two pupils in a Sukkur school learned the same poem in the same week. Rashid learned it the way poems are usually learned in his school: he recited it after the teacher, line by line, until he could produce it without a single error. Sana learned it at home from her grandmother, who first told her the story the poem tells, acted out the fisherman's arrival with a scarf, and only then read the words.

A week later the teacher checked. Rashid's recitation was letter-perfect, but when she asked him what the fisherman was doing in the second verse, he had no idea there was a fisherman. Sana's recitation stumbled twice, but she corrected the stumbles herself, because the words were hanging on a story she understood.

Both pupils learned. Only one of them learned a poem. The difference is the whole of this topic.



Explanation

The cognitive turn: learning as mental activity

Behaviourism measured what learners do. The **cognitive** view asks what learners *think*, and defines learning as change in thinking: new understandings, new connections, new ways of representing the world (Seifert & Sutton, 2009). The turn was a correction, not a rejection. Behaviourist mechanisms still run - Rashid's recitation is pure operant conditioning, rehearsed and praised - but they cannot tell the teacher whether the recitation carries any meaning.

The course guide names two theorists for this turn, and they divide the work neatly. Piaget explains the **structure** a learner brings: the schemas that receive new experience. Bruner explains the **route** teaching can take: the order from action to image to symbol.

Piaget: schemas, assimilation and accommodation

A **schema** is an organised packet of knowledge - "cat", "fractions", "how school days go". Learning, in Piaget's account, is the constant adjustment of schemas to experience, and it moves in two directions (Seifert & Sutton, 2009).

Assimilation fits new experience into an existing schema. The Class 1 pupil who calls every four-legged animal "billi" (cat) is assimilating: dogs, goats and cows all go into the cat-schema, because that is the schema he has. Assimilation is efficient and mostly silent; it is how a learner absorbs the ordinary day.

Accommodation changes the schema to fit the experience that will not go in. When the pupil finally separates dog from cat - a new category, a rebuilt schema - that is accommodation. Accommodation is the expensive operation: it is effortful, it is where misunderstanding and confusion live, and it is where real learning happens.

The two run together all day. The pupil who has a schema for "sharing sweets among cousins" and meets the fraction $\frac{1}{4}$ will assimilate it ("a quarter is my share when four of us share") - and that assimilation is the teacher's ally, because it is what the schema is *for*. The pupil who then meets $\frac{3}{4}$ and cannot file it anywhere is being asked for accommodation, and that is the moment the teaching must help.

Two consequences for a teacher. First, **a new topic lands on the schemas the pupils already have**, so the teacher's first move is to find out what those schemas are - the rote-learned poem landed on nothing and slid off. Second, **confusion is not failure**; it is the signature of accommodation under way. A class that is quietly comfortable is often a class being assimilated into its own existing ideas, and a class that is productively puzzled is often a class rebuilding one.

The misconception to clear now. "Piaget's stages mean a pupil cannot be taught a concept early." The stages (Topic 2.3) describe readiness, not permission. What the stage reading forbids is demanding the *symbolic* form of a concept before its concrete grounding - which is exactly what Bruner's route, next, supplies a way to avoid.

Bruner: representation, discovery and the spiral curriculum

Jerome Bruner's contribution is the teaching-side of the cognitive turn, in three ideas.

Three modes of representation. Learners can hold knowledge as **enactive** (knowing by doing - the halving of a roti), **iconic** (knowing by image - a drawing of the roti), and **symbolic** (knowing by sign - the written $\frac{1}{2}$). Bruner argued that any subject can be represented in all three, and that teaching should move through them in that order (Seifert & Sutton, 2009). Miss Shabana's roti lesson from Topic 2.4 is Bruner's route exactly: break the roti (enactive), draw the roti (iconic), write $\frac{1}{2}$ (symbolic).

Discovery learning. Bruner held that pupils learn best when they discover the structure of a subject for themselves, with guidance, rather than receive it as finished product. In practice this does not mean the teacher disappears; it means the teacher arranges the experience so the structure shows up - the sink-and-float basin, the unfair-sharing story that makes fractions necessary. The discovery is guided, or it does not happen.

The spiral curriculum. Because any idea can be represented at several levels, a curriculum can introduce an idea early in an intuitive form and return to it later in a more formal one. Fractions meet Class 2 as sharing food, Class 4 as rotis and diagrams, Class 6 as number operations. Nothing is wasted; the early round becomes the schema the later round accommodates.

the view	behaviourist	cognitive
what learning is	a change in observable behaviour	a change in thinking and understanding
what the teacher arranges	consequences that follow behaviour	experiences that meet and rebuild schemas
what counts as evidence	what the pupil can do, seen from outside	what the pupil can explain, transfer and use in a new situation
a classroom instance	Rashid's letter-perfect recitation	Sana's self-corrected recitation hanging on a story

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Activity: Two lessons, one order

In pairs, about 20 minutes.

Below is one concept - "why wet clothes dry" - and two lesson sketches.

Lesson A. The teacher writes "evaporation" on the board, defines it, states that heat makes water turn into vapour, and sets five questions whose answers are in the notes on the board.

Lesson B. The teacher hangs a wet cloth at the window at assembly and a second one in the store room, asks the class to predict which will dry first, records the predictions, and returns at home time to compare. The word "evaporation" appears only after the class has described what happened.

1. For each lesson, say where the enactive, iconic and symbolic representations appear - or note where one is missing.
2. For Lesson A, name what the pupils will most likely do with the definition (assimilate it into what schema, or fail to file it at all).

3. Decide: which lesson demands more from the teacher, and what exactly does it demand?

Check your understanding

1. Define schema in your own words, and give one example from your own schooling. (*Understand*)
2. Explain assimilation and accommodation with the "billi" example or one of your own, in two sentences. (*Understand*)
3. Rashid recited perfectly and understood nothing; Sana stumbled and understood. Explain the difference using this topic's terms. (*Apply*)
4. Name Bruner's three modes of representation, and place one lesson you have taught or observed into the order. (*Understand*)
5. A colleague says "discovery learning means the teacher does nothing." Give the corrected version in two sentences, using the word "guided". (*Understand*)

Summary

The cognitive turn defines learning as change in thinking. Piaget gives it a structure: schemas, adjusted by assimilation (new experience filed into what exists) and accommodation (the schema rebuilt), with confusion as the signature of the second. Bruner gives it a route: enactive before iconic before symbolic, discovery guided by the teacher's arrangement, and a spiral that returns to each idea in a stronger form. A new topic lands on the schemas the pupils already have; the teacher's first job is to find out what those are, and the second is to order the experience so the structure can show up.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can define schema, assimilation and accommodation with one example of each.
- ☐ I can explain why productive confusion is not failure.

- ☐ I can name Bruner's three modes and place a real lesson in their order.
- ☐ I can state what "guided" adds to discovery learning, and what the spiral curriculum assumes.

Try this at your practicum school

Take the next lesson you will teach and write its single most important new concept. Before teaching it, ask two pupils what they already think about it - their words, not the textbook's. Note whether your lesson's first activity meets their schemas or ignores them, and what you would change.

Summative task

Write **250-300 words** rebuilding Lesson A.

Using this topic's ideas, redesign the wet-clothes lesson so the concept of evaporation is built the way Sana's poem was built. Your answer must name: what the pupils already have (the schemas you will meet), where the enactive, iconic and symbolic moments sit, and where you expect accommodation - the productive confusion - to appear. Close with the one thing the redesigned lesson demands of you that Lesson A did not.

Mini-rubric

Criterion	What a strong answer shows
Use of the theory	Schemas, the three modes and accommodation are used accurately, not as decoration.
Order and evidence	The redesign runs enactive to iconic to symbolic with a concrete check at each step. (<i>Analyze</i>)
Honesty about cost	The extra demand on the teacher is named specifically.

Criterion	What a strong answer shows
Plain, accurate writing	Terms used correctly; no padding; within the word count.

Further reading

- Seifert, K., & Sutton, R. (2009). *Educational Psychology* (2nd ed.), Chapter 2. University of Manitoba. CC BY.
<https://open.umn.edu/opentextbooks/textbooks/educational-psychology>
- Vosniadou, S. (2001). *How children learn*. Educational Practices Series No. 7. International Academy of Education / UNESCO IBE. <https://eric.ed.gov/?id=ED455955>

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